Michael **Goodale**

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Education

PhD in Cognitive Science and Linguistics — Institut Jean Nicod, École normale supérieure

FUNDED BY ED474 AND THE LEARNING PLANET INSTITUTE AT UNIVERSITÉ PARIS-CITÉ — September 2023 — September 2026

Thesis: *BRaLL: Barely Rational Language Learners* supervised by Prof. Salvador Mascarenhas and Prof. Yair Lakretz

Masters in Cognitive Science — École normale supérieure

SPECIALISATION IN LINGUISTICS — September 2020 — May 2022

- GPA of 17.3/20 (*Mention très bien*)
- Thesis: *Manifolds as Conceptual Representations in Natural Language Semantics* supervised by Prof. Salvador Mascarenhas

B.A. & Sc. Honours — McGill University

MAJOR IN COGNITIVE SCIENCE WITH FOCUS IN LINGUISTICS AND MINOR IN COMPUTER SCIENCE — September 2015 — May 2019

- Graduated with First Class Honours and a GPA of 3.90/4.00
- Honours Thesis: *Probabilistically Reconstructing the Allophonic Rules of English Stops} supervised by Prof. Morgan Sonderegger*

Experience

Ingénieur des études (Research engineer) — École Normale Supérieure

September 2022 — 2023

Funded by a PRAIRIE Research Grant, studying how theories of pragmatics influence inferences made by NLI models and analysing the Bayesian models of language acquisition with Prof. Salvador Mascarenhas

Research intern — École Normale Supérieure

2021-2023

Analysed adjective representations in large language models leading to a poster at BlackboxNLP at EMNLP 2021 and an oral presentation at the Bridges and Gaps workshop at ESSLII 2022 with Prof. Salvador Mascarenhas

Research intern — École Normale Supérieure

September 2020—December 2020

Exploratory analysis of the learnt phonological representations of contrastive predictive coding and other unsupervised models of sound as part of the Cognitive Machine Learning team

Research Developer Intern — Nuance Communications

September 2019—December 2020

Natural Language Processing tasks involving BERT, Neural Machine Translation, and semantic similarity on large quantities of scraped text. Created a pipeline for training new languages on BERT in a specific semantic domain by creating new datasets automatically which saved significant investment in purchasing new datasets

Research Assistant — McGill University

May 2018—September 2019

Research with Prof. Morgan Sonderegger in phonetics, phonology and dialectology with web programming in Python using Django and front-end development with React as part of the SPADE ISCAN project

Research Assistant — Cambridge Brain Sciences

May 2017—September 2017

Exploratory data analysis of neurocognitive tests with Python, web programming in Ruby

Research Assistant — University of Western Ontario

May 2016—September 2017
May 2017–September 2017

Research with Prof. Marc Joanisse, running psycholinguistic studies, preparing stimuli with E-Prime, and analysis of fMRI data

Teaching Experience

Teaching Assistant — École normale supérieure

September 2023 — January 2024
September 2024 — January 2025

Introductory linguistics for first year masters students at École normale supérieure. Responsibilities included teaching two weekly tutorials and grading assignments as well as a guest lecture on computational language modelling.

Tutor for TalENS Paris

September 2021 — December 2021

Teaching weekend classes in French at the undergraduate level about different sub-fields of linguistics to high school students from disadvantaged schools as part of the TalENS program

Skills

Programming Languages	**Rust**, **Python**, JavaScript, Bash, C, LaTeX, Typst
Programming libraries	**PyTorch**, **NumPy**, **Pandas**, MatplotLib, Seaborn, HuggingFace Transformers, Docker
Spoken Languages	**English**, French (C2)

Side projects

- **Minimalist Grammar Parser**: A parser and string generator for Minimalist Grammars (Stabler, 1997) written in Rust which can also handle semantic composition in a lambda calculus defined over the lexicon. Includes extensions for use in Python, Javascript and Typst
- **Bekos**: A Bayesian Learner for Probabilistic Languages of Thought in Rust. Using Monte-Carlo Markov Chains, this library allows the learning of basic LISP-like programs from limited data to model human learning in a symbolic fashion
- **Prefrontal Cortex as Meta-Reinforcement Learning re-implementation** Re-implementation of *Prefrontal cortex as a meta-reinforcement learning system* (Wang et al, 2019) in PyTorch with new 2D maze environments.
- **Wall Street Bets analyser** Financial backtesting algorithm using sentiment analysis of comments from the /r/WallStreeetBets subreddit. Bull-bearish indicator built with BERT and the HuggingFace Transformers library, Named Entity Recognition to find references to specific stocks built with SpaCy and backtesting with the Backtrader framework

Honours

- **Fire PhD Grant** 2023-2026
- **PRAIRIE AI Research Grant in Social Science and the Humanities** 2022-2023
- **Bourse Frontenac** Master's research award from the Fonds de recherche du Québec – Nature et Technologies 2020
- **Dean's Honours List** Awarded to top 10% of the degree upon graduation 2019
- **McGill Department of Linguistics Award for Excellence in Research** 2019
- **Arts Undergraduate Research Internship Award** Research award with Morgan Sonderegger at McGill University 2018
- **Top Three at McHacks (McGill Hackathon) out of 97 teams** 2017
- **NSERC USRA** Research award with Marc Joanisse at Western University 2017
- **Faculty of Science In-Course Scholarship** 2016
- **JW McConnell Scholarship** Renewable scholarship covering all tuition and related expenses for four years 2015-2019
- **The Governor General's Bronze Academic Medal** Highest graduating average from my high school 2015

Publications

- **Michael Goodale**, Salvador Mascarenhas, Yair Lakretz (2025) *Meta-learning neural mechanisms rather than Bayesian priors*. Accepted at ACL 2025 (Main) arXiv
- **Michael Goodale**, Salvador Mascarenhas (2024) *Fodor and Pylyshyn's systematicity challenge still stands: A reply to Lake and Baroni*. Oral Presentation. 3rd International Conference on Human and Artificial Rationalities.
- **Michael Goodale** (2023). *Sense as sampling propensity*. Poster. Semantics and Linguistic Theory (SALT) 34. LingBuzz,
- **Michael Goodale**, Salvador Mascarenhas (2023). *Fodor and Pylyshyn's systematicity challenge still stands: A reply to Lake and Baroni (2023)* LingBuzz
- **Michael Goodale** (2023, November 17). *Sense as sampling propensity*. 13th Paris Amsterdam Logic Meeting of Young Researchers.
- **Michael Goodale**, Salvador Mascarenhas (2023). *Systematic polysemy in adjective-noun combination in contextual word embeddings*. LingBuzz
- Can Konuk, **Michael Goodale**, Tadeg Quillien, Salvador Mascarenhas (2023). *Plural causes in causal judgement*. Proceedings of the Annual Meeting of the Cognitive Science Society, Sydney, Australia. PsyArXiv
- **Michael Goodale** & Salvador Mascarenhas (2022). *Do contextual word embeddings represent richly subsective adjectives more diversely than intersective adjectives?* Bridges and Gaps Workshop ESSLLI 2022, Galway, Ireland.
- Michael McAuliffe, Arlie Coles, **Michael Goodale**, Sarah Mihuc, Michael Wagner, Jane Stuart-Smith, and Morgan Sonderegger. (2019) ISCAN: a system for integrated phonetic analyses across speech corpora. Proceedings of the 19th International Congress of Phonetic Sciences.